CSE 4345: Big Data Analytics

Week 1, Part 1 — Digital Data Landscape & Introduction to Big Data

Department of Computer Science & Engineering
Premier University, Chittagong

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Course & Lecture Information

Course Details

Code: CSE 4345
Credits: 3 (Theory)
Type: Elective – Data Science
Pre-req: CSE 2221 – DBMS

Today

Week / Part: 1 of 11 | Part 1 of 2
CLO: CLO1
PLO: PLO(a)

CLO1

"Understand big data characteristics"

Part 1 Covers

- The modern data explosion
- Types of digital data
- The 3Vs and 5Vs framework
- Distributed computing basics
- Ivy League alignment
- Student assessment pool

Lecture Agenda

The Data Revolution

We Are Living in the Data Age

Data generated every 60 seconds (2024 estimates):

- **6,000** tweets posted per second on X (Twitter)
- **500 hours** of video uploaded to YouTube per minute
- **1 million** messages sent on WhatsApp
- **5 million** Google search queries executed
- **500 TB** of new data ingested by Facebook daily
- **50,000** apps downloaded from Apple App Store

IDC Global Datasphere Forecast

By 2025, the world will generate and replicate **175 zettabytes** of data annually—a 23× increase from 2018.

Data Volume Units

Unit	Bytes
Kilobyte (KB)	10^3
Megabyte (MB)	10^6
Gigabyte (GB)	10^9
Terabyte (TB)	10^{12}
Petabyte (PB)	10^{15}
Exabyte (EB)	10^{18}
Zettabyte (ZB)	10^{21}
Yottabyte (YB)	10^{24}

The Fundamental Problem: Single Machine vs. Reality

What a Single Server Can Handle (2024)

Resource	Maximum
RAM (high-end server)	$\sim$64 TB
Disk throughput (SSD)	$\sim$7 GB/s
RDBMS comfortable limit	$\sim$10–50 TB
Time to read 1 PB at 7 GB/s	$\approx$41 hours

The Gap

Modern organisations generate data at rates that **fundamentally outpace** what any single machine can store, process, or query in acceptable time.

The Big Data Engineering Response

- **Distribute** data across hundreds of commodity nodes
- **Move computation to data** rather than fetching data to a central processor
- **Replicate** blocks for fault tolerance (3$\times$ in HDFS)
- **Parallelise** every operation—divide, compute, merge

Core principle (GFS, 2003):
"Commodity hardware will fail. Design for failure from the start."

Types of Digital Data

Three Categories of Digital Data

Structured
RDBMS, CSV, Excel

~20%
of world data

Semi-Structured
JSON, XML, Logs

~10%
of world data

Unstructured
Images, Video, Free Text

~70%
of world data

Key Insight

Over **80%** of the world's data is unstructured—this is the frontier that Big Data tools are built to unlock. Traditional SQL databases cannot process it without significant pre-processing.

Structured Data

Definition

Data organised into a **predefined schema** (rows and columns). Schema is enforced *at write time* (**schema-on-write**).

Characteristics:

- Fixed, known schema before ingestion
- Stored in relational databases (MySQL, Oracle, PostgreSQL)
- Queryable directly with standard SQL
- Strong ACID guarantees
- Easy to index, filter, aggregate

Domain Examples

Sample: Bank Transaction Table

TxID	Date	Amt (BDT)	Type
T001	2024-01-01	5,000	CR
T002	2024-01-01	1,200	DR
T003	2024-01-02	8,750	CR

Big Data relevance:

Structured data in legacy RDBMS is often the *source* that must be *migrated* into Hadoop or a data lake via **Apache Sqoop** (covered in Week 8).

Semi-Structured Data

Definition

Data with a **flexible, self-describing schema** embedded within the data itself. No separate schema definition is required.

Characteristics:

- Schema carried by the data (tags, keys, attributes)
- Human-readable and machine-parseable
- Schema can evolve without breaking existing records
- Dominant format for web APIs and event logs

Domain Examples

REST API responses (JSON), configuration files

Sample: Twitter API JSON Response

```
{
  "user_id":"bd456789",
  "tweet":"Big data is transforming
    Bangladesh fintech!",
  "timestamp":"2024-01-15T10:32Z",
  "likes":142,
  "retweets":37,
  "location":"Chittagong,BD",
  "media":null
}
```

Big Data context: Apache Hive and Spark can query JSON natively using **schema-on-read**—no schema pre-definition required at ingest time (covered in Week 7).

Unstructured Data—The Big Data Frontier

Definition

Data with **no predefined schema or relational format**. Cannot be stored in row/column tables without significant transformation.

Major categories:

- **Text:** Social media posts, news articles, clinical notes, legal contracts
- **Images:** X-rays, satellite photos, product images, CCTV frames
- **Video:** YouTube, dashcam footage, surgical recordings
- **Audio:** Call-centre recordings, podcasts, ECG waveforms
- **Sensor streams:** GPS pings, IoT telemetry, EEG

Integrated Healthcare Record (Bangladesh Context)

A single outpatient visit generates all three types:

- **Structured:** Blood pressure, pulse, lab numeric results (RDBMS table)
- **Semi-structured:** HL7/FHIR XML discharge summary
- **Unstructured:** Physician's handwritten notes (scanned PDF), ultrasound DICOM image, voice prescription recording

All three types must be **integrated** to build a complete clinical picture—a core CLO2 challenge.

Data Types: Comparison Summary

Dimension	Structured	Semi-Structured	Unstructured
Schema	Predefined (rigid)	Self-describing (flexible)	None
Storage	RDBMS	Files, NoSQL, APIs	Object stores, HDFS
Query tool	SQL	XPath, JSONPath, Hive	NLP, CV, Spark MLlib
Schema timing	On-write	On-read	N/A
World share	$\sim$20%	$\sim$10%	$\sim$70%
Big Data tool	Sqoop $\rightarrow$ Hive	Kafka, Hive	Spark + ML models
Example	Bank transaction	Tweet JSON	MRI scan image

Exam Tip

JSON is semi-structured, not unstructured—a common mistake. JSON carries keys (a schema) embedded within the data; an MRI image carries no embedded query schema at all.

Big Data: The 3Vs and 5Vs Framework

Defining Big Data—Multiple Perspectives

Gartner / META Group Definition

"Big data is high-volume, high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation."

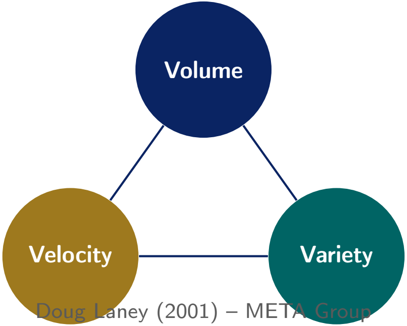
—Beyer & Laney, Gartner (2012)

McKinsey Global Institute (2011)

"Big data refers to datasets whose size is beyond the ability of typical database software tools to capture, store, manage and analyze."

Operational Definition

Big Data = data where the 3Vs exceed the capacity



Doug Laney (2001) – META Group

The 3Vs remain the **canonical definition** of Big Data, taught in every course from Stanford CS246 to CMU 15-721.

V1—Volume: The Scale of Data

Definition

The **sheer quantity** of data generated, stored, and processed. Volume is the most intuitive "big" in Big Data.

Production-scale examples:

- **Facebook / Meta:** 500 TB new data ingested per day
- **Google:** >100 PB processed per day (internal, 2023)
- **Walmart:** >2.5 PB customer transaction data
- **CERN LHC:** ≈15 PB of collision data per year
- **Bangladesh Election Commission:** 110 million NID biometric records
- **bKash:** Millions of micro-transactions per day

Textbook Reference

The Engineering Challenge

How do you store and query 500 TB when the largest single-instance database comfortably handles ≈50 TB?

Solution Architecture Preview

- **HDFS** (Week 3): Distribute data as 128 MB blocks across a cluster of thousands of commodity nodes
- **Replication factor 3**: Every block stored on 3 different nodes
- **Horizontal scaling**: Add nodes as data grows linearly—no downtime
- **Cloud object storage**: AWS S3, Google GCS for virtually unlimited scale

V2—Velocity: The Speed of Data

Definition

The **speed** at which data is generated, streamed, and must be processed or acted upon.

High-velocity real-world examples:

- **X (Twitter):** 6,000 tweets posted per second
- **Visa payment network:** 24,000 transactions/second (peak)
- **IoT sensor networks:** 1 billion device readings per hour
- **NYSE / DSE:** Millions of stock quotes per second
- **Pathao (BD):** GPS location ping every 3 s per active driver
- **Mobile banking (bKash):** Real-time fraud scoring in <200 ms

Processing Spectrum

Mode	Latency	Tool
Batch	Hours	Hadoop MR
Micro-batch	Seconds	Spark Streaming
Stream	Millisec	Apache Flink
Real-time	Sub-ms	Kafka + in-mem

Solution Preview

- **Apache Kafka** (Week 8): Distributed log for high-throughput event ingestion
- **Spark Streaming** (Week 6): Stateful micro-batch stream processing
- **Apache Flink**: True continuous stream

V3—Variety: The Diversity of Data

Definition

The **diversity** of data types, formats, sources, and update cadences that must be jointly ingested, integrated, and analysed.

Sources of variety:

- Multiple **formats**: CSV, JSON, XML, Avro, Parquet, DICOM, MP4
- Multiple **sources**: RDBMS, REST APIs, sensors, social media, ERP
- Multiple **languages** and character encodings (UTF-8, UTF-16, Bengali)
- Multiple **update frequencies**: historical snapshots + live streams
- Multiple **semantic meanings**: a “date” field could be

The Engineering Challenge

How do you JOIN a customer record from MySQL with their social media posts, call-centre audio transcription, and purchase-history CSV, all in one query?

Bangladesh Government Analytics Platform

Integrating data from 50 district hospitals requires:

- MySQL RDBMS tables (structured)
- Pharmacy sales CSV files (structured)
- Community health worker XML reports (semi-structured)

Extended Framework: The 5Vs of Big Data

V	Definition	Real-World Example
Volume	Scale of data stored and generated	Facebook: 500 TB/day ingested
Velocity	Speed of generation and required processing	Visa: 24,000 transactions per second
Variety	Diversity of formats, types and sources	CSV + JSON + DICOM + audio files
Veracity	Trustworthiness and accuracy of data	GPS signal drift, OCR errors, bot tweets, missing sensor readings
Value	Business/societal utility extracted from data	Revenue gain, lives saved, fraud prevented

Exam-Critical Distinction

The **original 3Vs** (Laney, 2001) addressed *systems* challenges. The **extended 5Vs** (Gartner, 2012)

Distributed Computing Fundamentals

Scaling Strategies: Vertical vs. Horizontal

Vertical Scaling (Scale Up)

Add more power to **one machine**: faster CPU, more RAM, larger disks.

- Simple—no distributed-systems complexity
- Hard physical ceiling (max RAM $\approx$ 64 TB today)
- Single point of failure
- **Exponentially expensive**: a $2\times$ more powerful server often costs $8\times$ more

Traditional RDBMS Approach

Oracle Exadata (vertical) can cost \$500,000+/year in licences alone.

Horizontal Scaling (Scale Out)

Add more **commodity machines** to the cluster.

- Each node is cheap (\$3,000–\$8,000 commodity server)
- Near-linear scalability: $2\times$ nodes $\approx$ $2\times$ throughput
- Fault tolerant: losing one node does not halt the cluster
- Google's GFS cluster (2003): **thousands** of commodity PCs, each expected to fail

Yahoo! Production Evidence (2008)

10,000 commodity nodes, 5 PB storage—cheaper than one equivalent mainframe, and far more

The CAP Theorem (Brewer, 2000)

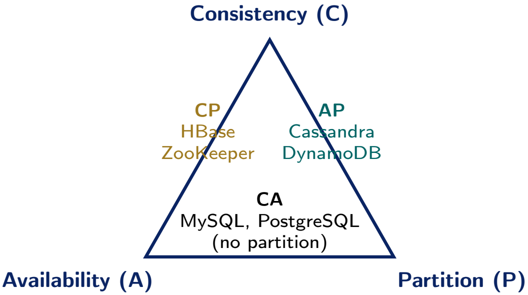
Theorem Statement (Brewer, PODC 2000; formally proved: Gilbert & Lynch, 2002)

A distributed data system can simultaneously guarantee **at most two** of:

- **C**onsistency: every read returns the most recent write
- **A**vailability: every request receives a response (not guaranteed to be latest)
- **P**artition tolerance: system operates despite network partitions

Practical Implication

In any real distributed system, network partitions *will* occur. P must be tolerated. Therefore, the design



Big Data Choices

HBase (CP): used in Hadoop—strong consistency, sacrifices availability on partition.
Cassandra (AP): used by Netflix—always available, eventual consistency.

Batch vs. Stream Processing

Dimension	Batch Processing	Stream Processing
Data state	Stored / historical	In-motion / live
Latency	Minutes to hours	Milliseconds to seconds
Throughput	Very high (optimised for)	Moderate
Fault recovery	Re-run the batch	Exactly-once semantics
Typical use case	Monthly billing, ML training, nightly ETL	Fraud detection, real-time alerts, live dashboards
Technologies	Hadoop MapReduce, Hive, Spark Batch	Apache Kafka, Spark Streaming, Apache Flink
Complexity	Lower	Higher

Week 8 Preview: Architectural Choice

The **Lambda architecture** runs both layers in parallel (batch for accuracy + stream for latency). The

Ivy League Alignment

Stanford CS246: Mining Massive Datasets

Course Philosophy (Leskovec, Rajaraman & Ullman)

"The challenge of big data is not just storage—it is running algorithms that are fast enough to be useful on datasets that do not fit in the memory of a single machine. This forces a rethinking of everything."

— *Mining of Massive Datasets*, Preface

Key Benchmark: Sub-Quadratic Requirement

- On $n = 10^9$ records, an $O(n^2)$ algorithm requires 10^{18} operations
- At 10^9 ops/second, that is **31.7 years** of compute time
- Big Data forces every algorithm to target $O(n)$ or $O(n \log n)$

Benchmarks Across Ivy League

- **Stanford CS246**: MapReduce, Spark, LSH, graph mining, recommender systems
- **Harvard CS265**: Storage systems, query optimisation, indexing at scale
- **MIT 6.830**: Distributed consistency, transaction models
- **CMU 15-721**: Modern columnar and in-memory databases
- **Columbia COMS 4995**: End-to-end big data engineering pipelines

CSE 4345 Positioning

This course covers the **foundational**

Student Assessment Pool

Assessment—Multiple Choice Questions: Set A *CLO1*

Instructions

Select the single best answer. Answers are **bolded** for instructor reference; present without bold to students.

Q1 Who first formally proposed the **3Vs** framework for Big Data?

(a) Jim Gray (b) **Doug Laney** (c) Jeffrey Dean (d) Erik Brynjolfsson

Q2 Which of the following is an example of **unstructured data**?

(a) SQL relational table (b) CSV file (c) **MRI scan image** (d) JSON API response

Instructor Notes

Q1: Students commonly confuse Laney with MapReduce authors (Dean & Ghemawat). **Q2:** JSON (option d) is a common wrong answer—it is *semi*-structured because it carries key names; an MRI has no schema at all.

Assessment—Multiple Choice Questions: Set B *CLO1*

Q3. The CAP theorem states that a distributed system cannot simultaneously guarantee more than two of:

- (a) Cost, Availability, Performance
- (b) **Consistency, Availability, Partition Tolerance**
- (c) Correctness, Atomicity, Persistence
- (d) Concurrency, Accuracy, Partitioning

Q4. Which "V" of Big Data specifically refers to the **speed** at which data is generated and must be processed?

- (a) Volume (b) Variety (c) **Velocity** (d) Veracity

Instructor Notes

Q3: All four options are designed to sound plausible—requires precise recall. **Q4:** "Veracity" (option d) is the most common wrong answer since it starts with "Ve-"; students must know the precise definitions of each V.

Assessment—True / False *CLO1*

Instructions

State True or False. Award full marks only if the student provides a **one-sentence justification**.

Statement	Answer
1. Structured data accounts for approximately 80% of all digital data in the world.	False
2. The 3Vs Big Data framework was extended to 5Vs by adding Veracity and Value.	**True**
3. Horizontal scaling means adding more RAM and faster CPUs to a single existing server.	False
4. Batch processing is suitable for detecting a stolen credit card being used in real time.	False

Assessment—Descriptive Questions

Instructions

Answer each question in **100–150 words**. Use specific terminology covered in class.

D1. Explain the **3Vs** (and the extended 5Vs) of Big Data, providing one concrete real-world example for *each* dimension.

D2. Differentiate between **structured, semi-structured, and unstructured data**. Provide examples exclusively from the **healthcare domain in Bangladesh**.

D3. State and explain the **CAP theorem**. Why is it a fundamental constraint for designing distributed Big Data systems? Give one concrete example of a real system that makes each of the three trade-offs (CA, CP, AP).

Assessment—Analytical Questions *CLO1*

Instructions

Answer in **200–300 words**. Show step-by-step reasoning. Partial credit is awarded for correct reasoning even with incomplete conclusions.

A1. **5Vs Analysis—Pathao (Bangladesh ride-sharing):**
Pathao collects a GPS ping every 3 seconds from 500,000 active drivers, integrates real-time traffic feeds, surge pricing algorithms, and customer written reviews.
- Analyse this scenario through **all 5Vs**.
- Which V poses the greatest **engineering challenge**? Justify.
- What happens to the overall system if Veracity is compromised (GPS signal drifts by 200 m)?

A2. **Processing Paradigm Trade-off:**
Compare batch and stream processing for: **(a)** generating a monthly bank statement, **(b)** detecting that a stolen bKash mobile account is being drained 10 seconds after theft. Justify why each scenario demands a different processing model and identify the key engineering trade-offs.

Textbook & Reading References

Primary Textbooks for Week 1

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Chapter 1**
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Chapters 1, 3**
- **[MMS]** Mayer-Schönberger & Cukier. *Big Data: A Revolution*, Houghton Mifflin, 2013. **Chapter 1**
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed., Cambridge, 2020. **Chapter 1**
  Free PDF: `mmds.org`

Supplementary—Covered in Week 1 Part 2 (Seminal Papers)

- Laney, D. (2001). *3D Data Management: Controlling Data Volume, Velocity and Variety*. META Group Research Note. **[PRIMARY]**
- Brewer, E. (2000). *Towards Robust Distributed Systems*. PODC Keynote.
- Gilbert & Lynch (2002). *Brewer's Conjecture and the Feasibility of Consistent Available Partition-Tolerant Web Services*. SIGACT.

Questions & Discussion

Week 1, Part 1 | CSE 4345 Big Data Analytics

"Without data you're just another person with an opinion."

—W. Edwards Deming

Next Lecture: Week 1, Part 2

Seminal Research: Laney (2001) · CAP Theorem Formal Proof
Case Study: McKinsey Global Institute (2011) · Live Data Exercise